TrueFalse Challenge: Solution

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1 Solutions

The value of n is: $\boxed{895001500}$

1. $\boxed{\textit{False.}}$ We can see that, if $P = Q = R = \emptyset$, then we know $P \cup Q \cup R = P \cap Q \cap R = \emptyset$. However, this is not true the other way around. The union of the three sets will contain all values in any of the sets, and the intersection of the three will contain elements common to all three. Thus, if $P = Q = R$, then $P \cup Q \cup R = P \cap Q \cap R = P = Q = R$, since each of the values are contained in each set, and no set contains more than the common values. For example, consider $P = Q = R = \{1, 2\}$. Then $P \cup Q \cup R = \{1, 2\}$ (all values in any set) and $P \cap Q \cap R = \{1, 2\}$ (all values common to all three sets). We can see that $P \cap Q \cap R = \emptyset$ is one case of this condition.

2. $\boxed{\textit{True.}}$ Let $a = x^2$, and note that we must have $a \geq 0$ since $x^2 \geq 0$. Then, we have

$$f(x) = \frac{1 - a^{50}}{1 + a} \qquad\qquad g(x) = \frac{1 + a^{50}}{1 - a}$$

We can prove that, for any positive rational number $k = \frac{a}{b}$ and nonnegative real numbers x and y, that

$$\frac{a - x}{b + y} \leq \frac{a}{b} \leq \frac{a + x}{b - y}.$$

Intuitively, when we decrease the numerator and increase the denominator, both of those actions decrease the value of a rational number, and increasing the numerator and decreasing the denominator increases the value. We can prove the first statement by making a series of reversible steps that result in a statement that is always true.

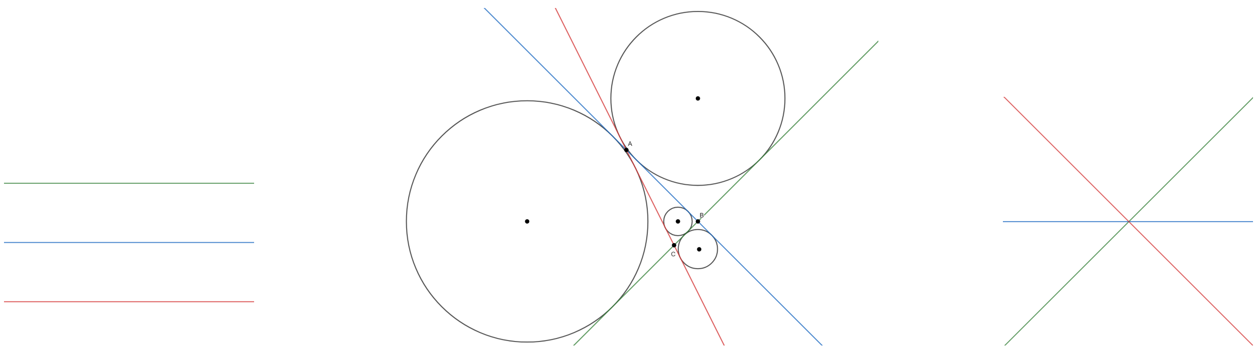
$$\begin{aligned} \frac{a - x}{b + y} &\leq \frac{a}{b} \\ b(a - x) &\leq a(b + y) \\ ab - bx &\leq ab + ay \\ -bx &\leq ay \end{aligned}$$

Which is always true since a, b are positive and x, y are nonnegative. The proof for $\frac{a}{b} < \frac{a+x}{b-y}$ follows by a similar method. If $k < 0$, the statement is not always true, but that case isn't relevant to this question. For all x, a and a^{50} are nonnegative, and we have

$$f(x) = \frac{1 - a^{50}}{1 + a} \leq \frac{1}{1} \leq \frac{1 + a^{50}}{1 - a} = g(x).$$

We have found an upper bound on $f(x)$ and a lower bound on $g(x)$, and since these are both achievable at $x = 0$, we have found the maximum of $f(x)$ and the minimum of $g(x)$ to be $\frac{1}{1} = 1$, so the statement is true.

3. $\boxed{\textit{False.}}$ To have a circle tangent to three distinct lines in a plane, there needs to exist at least one section of the plane that is bounded by all three lines. For example, for the lines $y = x$, $y = -x$, and $x = 1$, the section to the right of the origin is bounded by all three lines and so we can find a tangent circle (in this case there are others too!). However, this isn't true for any set of three lines. No circle can be tangent to any three concurrent lines, since there is no area bounded by all three lines. Another contradictory case is three distinct parallel lines. There are no circles tangent to all three tangent lines, because a tangent circle has to lie strictly on one side of a line, which means any circle tangent to one of the parallel lines cannot be tangent to both of the other two.



(a) Parallel lines, no tangent circles

(b) Three nonconcurrent lines and tangent circles

(c) Concurrent lines, no tangent circles

Problem 3: Tangent circles to three distinct lines (Would appreciate if this could be remade, they dont look very professional)

4. $\boxed{\textit{True.}}$ The statement is analogous to the Hockey Stick Identity. We can see for small values that this works, for example with $n = 5$ and $k = 3$:

$$\binom{3}{3} + \binom{4}{3} + \binom{5}{3} = 1 + 4 + 10 = 15 = \binom{6}{4}.$$

The general statement can be proven with induction. For the base case, take $n = k$:

$$\sum_{i=k}^{k} \binom{i}{k} = \binom{k}{k} = 1 = \binom{k+1}{k+1} = \binom{n+1}{k+1}.$$

Now, we assume that the statement is true for some $n = m$, where $m \geq 1$ is an integer, and we want to show that this implies the statement is true for $n = m+1$. We have by assumption that

$$\sum_{i=k}^{m} \binom{i}{k} = \binom{m+1}{k+1},$$

and we want to show this implies

$$\sum_{i=k}^{m+1} \binom{i}{k} = \binom{m+2}{k+1}.$$

We can take the last term from this sum and get

$$\sum_{i=k}^{m+1} \binom{i}{k} = \left(\sum_{i=k}^{m} \binom{i}{k}\right) + \binom{m+1}{k} = \binom{m+1}{k+1} + \binom{m+1}{k},$$

with the last step due to the inductive assumption. From Pascal's Identity, $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$, so we can rewrite

$$\binom{m+1}{k+1} + \binom{m+1}{k} = \binom{m+2}{k+1},$$

which completes the induction.

5. $\boxed{\textit{True.}}$ If k is rational, $\cos\left(\frac{2\pi n}{k}\right)$ will repeat values, and if k is irrational, it won't. First, let k be a rational number. Then, we want to show that $\cos\left(\frac{2\pi n}{k}\right)$ repeats over positive integers n. This means, for two distinct positive integers q and r, we must have

$$\cos\left(\frac{2\pi q}{k}\right) = \cos\left(\frac{2\pi r}{k}\right).$$

The full condition for $\cos(x) = \cos(y)$ is if $x = \pm y + 2\pi d$ for some integer d, since $\cos(x)$ is an even function so $\cos(x) = \cos(-x)$. We have that, for some integer m,

$$\begin{aligned}
\frac{2\pi q}{k} &= \pm\frac{2\pi r}{k} + 2\pi m \\
\frac{q}{k} &= \pm\frac{kr}{k} + m \\
\frac{1}{k}(q \mp r) &= m \\
q \mp r &= km
\end{aligned}$$

Since k is rational, there are infinitely many integers m such that km is an integer, then we can find (q, r) such that the equation is true. For example, let $k = \frac{3}{4}$, then $q = 4$ and $r = 1$ gets the solution

$$\cos\left(\frac{32\pi}{3}\right) = \cos\left(\frac{8\pi}{3}\right)$$

Which is true since $\frac{32\pi}{3} = \frac{8\pi}{3} + 8\pi$ (meaning $m = 4$, which agrees with what we had before). We have shown that, if k is rational, $\cos\left(\frac{2\pi n}{k}\right)$ repeats value over positive integers n. However, if k is irrational, this doesn't hold. Let k be any irrational number. Then, we want to show if there exists distinct integers q and r such that

$$\cos\left(\frac{2\pi q}{k}\right) = \cos\left(\frac{2\pi r}{k}\right).$$

Using the same condition that $\cos(x) = \cos(y)$ if and only if $x = \pm y + 2\pi d$ for some integer d, we must have, for some integer m,

$$\begin{aligned}
\frac{2\pi q}{k} &= \pm\frac{2\pi r}{k} + 2\pi m \\
\frac{q}{k} &= \pm\frac{r}{k} + m \\
\frac{1}{k}(q \mp r) &= m \\
q \mp r &= km
\end{aligned}$$

Since q and r are integers, $q \mp r$ must also be an integer, and since k is irrational and m is an integer then km must be irrational. It is not possible for an integer to be irrational, so there is no possible way for $\cos\left(\frac{2\pi q}{k}\right) = \cos\left(\frac{2\pi r}{k}\right)$ if k is irrational. Thus, the statement is true, since $\cos\left(\frac{2\pi n}{k}\right)$ will never repeat a value over its range when k is irrational.

6. $\boxed{\textit{False.}}$ First, we attempt to find a general form for $g(a, b, p)$. We can notice a factoring similar to a geometric series:

$$\begin{aligned}\frac{a^p + b^p}{a + b} &= \frac{a^{p-1} + \frac{b^p}{a}}{1 + \frac{b}{a}} \\ &= a^{p-1} \frac{1 + (\frac{b}{a})^p}{1 + \frac{b}{a}} \\ &= a^{p-1} - a^{p-2}b + a^{p-3}b^2 + \cdots - ab^{p-2} + b^{p-1}.\end{aligned}$$

We can use the Euclidean Algorithm by first evaluating the expression $\pmod{a + b}$. Notice that $a \equiv -b \pmod{a + b}$ and that $p - 1$ is even.

$$(-b)^{p-1} - (-b)^{p-2}b + (-b)^{p-3}b^2 + \cdots + b^{p-1} \equiv (-1)^{p-1}pb^{p-1} \equiv pb^{p-1} \pmod{a + b}$$

By the Euclidean Algorithm, we simply get:

$$\gcd\left(\frac{a^p + b^p}{a + b}, a + b\right) = \gcd(pb^{p-1}, a + b)$$

We also know that $\gcd(a, b) = 1$, so $\gcd(b, a + b) = 1$, letting us further simplify:

$$\gcd(pb^{p-1}, a + b) = \gcd(p, a + b)$$

We now have a way to relate the quantity $a + b$ and the the odd prime p. We know that, if $p \mid (a + b)$, then $\gcd(p, a + b) = p$. Otherwise, we have $\gcd(p, a + b) = 1$.

Next, we can factor $f(x)$ with the following manipulation:

$$\begin{aligned}f(x) &= x^2 - 8x + 7 + 7 \\ f(x) &= (x - 1)(x - 7) + 7\end{aligned}$$

Since we are given the condition that $f(x_1) = f(x_2) = p$, we must consider values for which $f(x)$ attains a prime value. Notice that $f(7) = f(1) = 7$ from our factoring, which gives us equations $a_1 + b_1 = 7$ and $a_2 + b_2 = 1$. Since, for $p = 7$, we have $a_1 + b_1 = p$, then $\gcd(a_1 + b_1, p) = 7$, while $\gcd(a_2 + b_2, p) = 1 \neq 7$. Since we have found a counterexample to the statement, it must be false.

7. $\boxed{\textit{True.}}$ Figure 2 is the construction of the diagram by the problem statement. We want to show that all O_i lie on one of the two dotted rays. Note that all triangles $\triangle ABX_1$, $\triangle BX_1X_2$,$\triangle X_nX_{n+1}X_{n+2}$ are isosceles right triangles, so they are all similar to eachother. The key observation is that we have two dilations(or homotheties) with respect to point C.

   First, we want to show that the triangles $\triangle ABX_1$ and $\triangle X_iX_{i+1}X_{i+2}$ (where i is an odd positive integer) are images of eachother under a homothety with center C. Notice that corresponding vertices lie on the same ray: X_i and A, as well as X_{i+2} and X_i, lie on $\overrightarrow{AC}$, and X_{i+1} and B lie on $\overrightarrow{CB}$. Next, there must be a fixed ratio of dilation between any two triangles in the sequence. Consider $\triangle ABX_1$ and $\triangle X_1X_2X_3$. Altitude X_1X_2 bisects BC, so

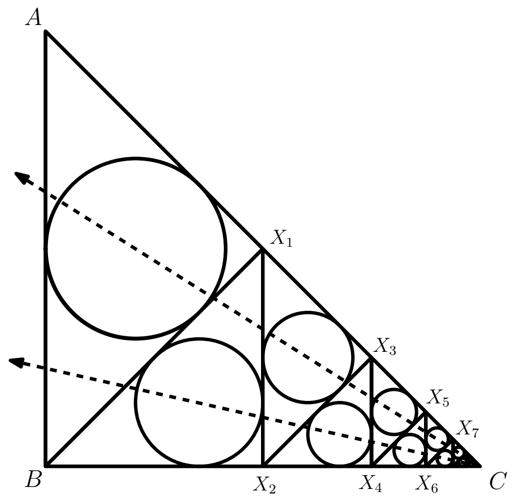


Figure 2

$\frac{CX_2}{BC} = \frac{1}{2}$, and $\triangle ABC \sim \triangle X_1X_2C$, so $\frac{X_1X_2}{AB} = \frac{1}{2}$ as well. Since both $\triangle ABX_1$ and $\triangle X_1X_2X_3$ are isosceles right triangles, they must be similar, and we have shown this similarity is with factor $\frac{1}{2}$. This can be done with any pair of triangles $\triangle X_iX_{i+1}X_{i+2}$ and $\triangle X_jX_{j+1}X_{j+2}$, where i and j are odd integers, and get a constant scale factor of $\frac{1}{2^{i-j}}$.

Similar steps will show that $\triangle BX_1X_2$ and $\triangle X_iX_{i+1}X_{i+2}$, for positive *even* integers i, are images of eachother under homothety centered at C. Under dilation, corresponding points are collinear with the center of dilation. The incenter of a triangle corresponds to the incenter of the image of that triangle, so for any dilation of a triangle, the incenter is collinear with the incenter of the image and the center of dilation. Since we have two separate dilations, we have two lines that the incenters will lie on. For integer $k > 0$, all O_{2k-1} lie on triangles of the form $\triangle X_{2k-1}X_{2k}X_{2k+1}$ or $\triangle ABX_1$, which we have shown to all be dilations of eachother, thus all O_{2k-1} are collinear since they correspond under dilation. Similarly, all O_{2k} correspond under dilation, and both of these dilations have center C, so the lines formed by these two sets of incenters must intersect at C.

8. $\boxed{\textit{True.}}$ We can represent both $\arccos(\cos x)$ and $\lim\limits_{n\to\infty} f^n(x)$ as piecewise functions. First, $\arccos(x)$ returns values between 0 and π, taking inputs from -1 to 1. $\arccos(\cos x)$ extends this range to $\mathbb{R}$, since the domain of $\cos(x)$ is $\mathbb{R}$ with range $[-1, 1]$. Since $\arccos(x)$ only outputs positive values, $\arccos(\cos x)$ will not return x when $x < 0$, rather, it returns $-x$. For all inputs x, we have $\cos(x) = \cos(\theta)$, for some $-\pi < \theta < \pi$, since $\cos(\theta)$ achieves the entire range of $[-1, 1]$ on that interval. So, we have

$$\arccos(\cos\theta) = \left\{ \begin{array}{ll} \theta & 0 \le \theta \le \pi \\ -\theta & -\pi \le \theta \le 0 \end{array} \right\}$$

Now, we can extend this to all real x. Each x corresponds to some θ such that $-\pi < \theta < \pi$ and $\cos(x) = \cos(\theta)$, and this implies that $x = \theta + 2\pi k$ for some integer k. Now, we can rewrite the piecewise function, for all integers k, as

$$\arccos(\cos x) = \left\{ \begin{array}{ll} x - 2\pi k & 2\pi k \le x \le 2\pi(k+1) \\ -x + 2\pi k & 2\pi(k-1) \le x \le 2\pi k \end{array} \right\}$$

Now for $\lim_{n\to\infty} f^n(x)$. Consider $g(x) = |x|$, and we are repeatedly applying the transformation $F(x) = ||x - \pi| - \pi|$ onto it. What does $F(x)$ do to a function? Well, first it translates the function π units down, reflects everything under the y axis back above it, then repeats this. We know that

$$g(x) = |x| = \left\{ \begin{array}{ll} x & x \geq 0 \\ -x & x < 0 \end{array} \right\}$$

Applying $F(g(x))$ (which is $f(x)$), we now have

$$F(g(x)) = \left\{ \begin{array}{lr} -x - 2\pi & x \in (-\infty, -2\pi] \\ x + 2\pi & x \in [-2\pi, -\pi] \\ -x & x \in [-\pi, 0] \\ x & x \in [0, \pi] \\ -x + 2\pi & x \in [\pi, 2\pi] \\ x - 2\pi & x \in [2\pi, \infty) \end{array} \right\}$$

Repeated application of $F(x)$ to $g(x)$ will make $F^n(g(x))$ alternate between $x - 2\pi k$ and $-x + 2\pi k$ (for changing k) over intervals of length π. Applying $F(x)$ infinitely allows this pattern to continue indefinitely in both directions, so there will be no interval $(-\infty, -2\pi k]$ or $[2\pi k, \infty)$ in the piecewise representation of $F^k(g(x))$ as k tends to infinity.

$$F^k(g(x)) = \left\{ \begin{array}{lr} -x - 2\pi k & x \in (-\infty, -2\pi k] \\ x + 2\pi k & x \in [-2\pi k, -2\pi(k-1)] \\ -x - 2\pi(k-1) & x \in [-2\pi(k-1), -2\pi(k-2)] \\ \vdots & \vdots \\ x - 2\pi(k-1) & x \in [2\pi(k-2), 2\pi(k-1)] \\ -x + 2\pi k & x \in [2\pi(k-1), 2\pi k] \\ x - 2\pi k & x \in [2\pi k, \infty) \end{array} \right\}$$

We can see that, comparing $F^k(g(x))$ and $F^{k+1}(g(x))$, they are identical on the range $[-2\pi k, 2\pi k]$, and this is true for all k. As $k \to \infty$, $-2\pi k \to -\infty$ and $2\pi k \to \infty$, so as long as $\lim_{n\to\infty} F^n(g(x))$ exists, it is reasonable to say that $\lim_{n\to\infty} F^n(g(x)) = \arccos(\cos x)$ as their piecewise representations will be identical. Notice that, if we had $F(x) = |x - \pi|$, this would not be true since $F^{2k}(g(0)) = 0$ while $F^{2k+1}(g(0)) = \pi$, so the function would alternate based on the parity of how many times F is applied and therefore the limit would not exist. However, since $F(x) = ||x-\pi|-\pi|$, the function does not alternate (the parity of applications of $|x - \pi|$ is always even), and instead converges to the piecewise function

$$\lim_{n\to\infty} F^n(g(x)) = \left\{ \begin{array}{ll} x - 2\pi k & 2\pi k \leq x \leq 2\pi(k+1) \\ -x + 2\pi k & 2\pi(k-1) \leq x \leq 2\pi k \end{array} \right\} = \arccos(\cos x).$$

We know that $\lim_{n\to\infty} f^n(x) = \lim_{n\to\infty} F^n(g(x))$, since the only difference is that $f(x)$ will have extra absolute value signs every application (for example, $f^2(x) = ||||||x| - \pi| - \pi|| - \pi| - \pi|$ which is the same as $F^2(g(x)) = |||||x| - \pi| - \pi| - \pi| - \pi|$). So, $\lim_{n\to\infty} f^n(x) = \lim_{n\to\infty} F^n(g(x)) = \arccos(\cos x)$, meaning the statement is true.

9. $\boxed{\textit{False.}}$ We are trying to figure out if the integral

$$\int_1^\infty \left(\frac{\lceil x \rceil}{\lfloor x \rfloor} - \frac{\lfloor x \rfloor}{\lceil x \rceil} \right) dx$$

converges or diverges. This setup is pretty intimidating, but instead of an integral we can consider an infinite sum. We can see that, if x is an integer, then $\lceil x \rceil = \lfloor x \rfloor = x$, so $\frac{\lceil x \rceil}{\lfloor x \rfloor} - \frac{\lfloor x \rfloor}{\lceil x \rceil} = 1 - 1 = 0$. Otherwise, the quantities of $\lceil x \rceil$ and $\lfloor x \rfloor$ do not change when x is between two integer values. So, when we integrate over a range of real numbers, $f(x) - \frac{1}{f(x)}$ will be constant for each interval between two integers (which is a range of length 1). Then, finding the area on this range would be a sum of rectangles of width 1 and height $f(x) - \frac{1}{f(x)}$. Since $\lceil x \rceil$ and $\lfloor x \rfloor$ always return integers, we can convert the integral into a discrete sum of rectangle areas.

Let n be the integer such that $n < x < n+1$ (we do not consider when x is an integer because it adds nothing to the area). Then, $\lfloor x \rfloor = n$ and $\lceil x \rceil = n+1$. We start at $n = 1$ (from the integral bounds), so the sum becomes

$$\sum_{n=1}^{\infty} \left(\frac{n+1}{n} - \frac{n}{n+1} \right) = \sum_{n=1}^{\infty} \left(1 + \frac{1}{n} - (1 - \frac{1}{n+1}) \right) = \sum_{n=1}^{\infty} \left(\frac{1}{n} + \frac{1}{n+1} \right).$$

We can rewrite this in terms of the harmonic series:

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{n+1} \right) = \sum_{n=1}^{\infty} \left(\frac{1}{n} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{n} \right) - 1 = -1 + 2 \sum_{n=1}^{\infty} \left(\frac{1}{n} \right).$$

This sum must diverge since the harmonic series is divergent, and since this discrete sum is equivalent to the given integral, the integral diverges as well, making the statement false.

10. $\boxed{\textit{True.}}$ We are tasked to compute

$$\int_1^{\infty} \int_1^{\infty} \frac{1}{xy(x^2 + y^2)} dx dy.$$

The statement asks if this is equivalent to the value Z that satisfies $e^{nZ} = m$, which is $\frac{\ln m}{n}$ for some positive integers n and m. We will deal with the inner integral first, that is

$$\int_1^{\infty} \frac{1}{xy(x^2 + y^2)} dx.$$

When integrating multivariable functions with respect to one variable, all other variables are considered constant, since we are only changing x. So, we will deal with y as a constant, which means it does not depend on x. So, we can write

$$\int_1^{\infty} \frac{1}{xy(x^2 + y^2)} dx = \frac{1}{y} \int_1^{\infty} \frac{1}{x(x^2 + y^2)} dx.$$

Now, we will solve for the indefinite integral of the integrand. Let $u = x^2 + y^2$, then $du = 2x dx$ (remember y is constant). We can write $dx = \frac{du}{2x}$ and $x^2 = u - y^2$ to get

$$\int \frac{1}{x(x^2 + y^2)} dx = \int \frac{1}{xu} \frac{du}{2x} = \int \frac{1}{2x^2 u} du = \frac{1}{2} \int \frac{1}{(u - y^2)(u)} du.$$

Now, we can use partial fraction decomposition, solving for A and B given

$$\frac{1}{(u - y^2)(u)} = \frac{A}{u} + \frac{B}{u - y^2}.$$

This gets

$$A(u-y^2)+Bu=1$$
$$u(A+B)-Ay^2=1$$

A and B must be constants since the denominator is linear in both terms. This means we must have

$$A+B=0$$
$$-Ay^2=1$$

Solving, we get $A=-\frac{1}{y^2}$ and $B=\frac{1}{y^2}$. So, the integral becomes

$$\frac{1}{2}\int\left(-\frac{1}{uy^2}+\frac{1}{y^2(u-y^2)}\right)du=\frac{1}{2y^2}\int\left(\frac{1}{(u-y^2)}-\frac{1}{u}\right)du.$$

Using the fact that $\int\frac{1}{x}dx=\ln|x|$, we find the indefinite integral to be

$$\frac{1}{2y^2}\int\left(\frac{1}{(u-y^2)}-\frac{1}{u}\right)du=\frac{\ln|u-y^2|-\ln|u|}{2y^2}+C=\frac{\ln|\frac{u-y^2}{u}|}{2y^2}+C=\frac{\ln|1-\frac{y^2}{u}|}{2y^2}+C=\frac{\ln|1-\frac{y^2}{x^2+y^2}|}{2y^2}+C.$$

Now, we can evaluate the definite integral. We have

$$\int_1^\infty\frac{1}{x(x^2+y^2)}dx=\left[\frac{\ln|1-\frac{y^2}{x^2+y^2}|}{2y^2}+C\right]_{x=1}^{x=\infty}=\lim_{N\to\infty}\left(\frac{\ln|1-\frac{y^2}{N^2+y^2}|}{2y^2}+C\right)-\left(\frac{\ln|1-\frac{y^2}{1+y^2}|}{2y^2}+C\right)$$

As $N\to\infty$, $\dfrac{y^2}{N^2+y^2}\to 0$, so $\ln|1-\frac{y^2}{1+y^2}|$ approaches $\ln(1)=0$, and the constants C cancel out. Then,

$$-\frac{\ln|1-\frac{y^2}{1+y^2}|}{2y^2}=-\frac{\ln|1-\left(1-\frac{1}{y^2+1}\right)|}{2y^2}=\frac{-\ln\frac{1}{y^2+1}}{2y^2}=\frac{\ln|y^2+1|}{2y^2}.$$

Now, plugging this back in to the original integral, we get

$$\int_1^\infty\frac{1}{y}\int_1^\infty\frac{1}{x(x^2+y^2)}dxdy.=\int_1^\infty\frac{1}{y}\frac{\ln|y^2+1|}{2y^2}dy=\int_1^\infty\frac{\ln|y^2+1|}{2y^3}dy=\frac{1}{2}\int_1^\infty\frac{\ln|y^2+1|}{y^3}dy.$$

We will finish this off with integration by parts. Let $u=\ln|y^2+1|$ and $dv=\dfrac{dy}{y^3}$. Then, $du=\dfrac{2y}{y^2+1}dy$ and $v=-\dfrac{1}{2y^2}$. Then, by integration by parts,

$$\int\frac{\ln|y^2+1|}{y^3}dy=-\frac{\ln|y^2+1|}{2y^2}-\int-\frac{1}{2y^2}\frac{2y}{y^2+1}dy=-\frac{\ln|y^2+1|}{2y^2}+\int\frac{1}{y(y^2+1)}dy.$$

We can solve for the second integral with partial fraction decomposition once again:

$$\frac{1}{y(y^2+1)}=\frac{A}{y}+\frac{B}{y^2+1}$$

From this, we get

$$
\begin{aligned}
A(y^2+1)+By &= 1 \\
Ay^2+By+A &= 1
\end{aligned}
$$

To match the constants, we take $A = 1$, which gets $B = -y$, so we have $\frac{1}{y(y^2+1)} = \frac{1}{y} - \frac{y}{y^2+1}$, then

$$\int \frac{1}{y(y^2+1)}dy = \int \left(\frac{1}{y} - \frac{y}{y^2+1}\right) dy = \ln|y| - \frac{1}{2}\ln|y^2+1| + C = \ln\left|\frac{y}{\sqrt{y^2+1}}\right| + C.$$

Now we can solve for the definite integral, getting

$$
\begin{aligned}
\int_1^{\infty} \frac{\ln|y^2+1|}{y^3}dy &= \left[-\frac{\ln|y^2+1|}{2y^2} + \ln\left|\frac{y}{\sqrt{y^2+1}}\right| + C\right]_{y=1}^{y=\infty} \\
&= \lim_{N\to\infty}\left(-\frac{\ln|N^2+1|}{2N^2} + \ln\left|\frac{N}{\sqrt{N^2+1}}\right|\right) - \left(-\frac{\ln|2|}{2} + \ln\left|\frac{1}{\sqrt{2}}\right|\right) \\
&= \lim_{N\to\infty}\left(-\frac{\ln|N^2+1|}{2N^2} + \ln\left|\frac{N}{\sqrt{N^2+1}}\right|\right) + \frac{\ln|2|}{2} - \frac{1}{2}\ln\left|\frac{1}{2}\right| \\
&= \lim_{N\to\infty}\left(-\frac{\ln|N^2+1|}{2N^2} + \ln\left|\frac{N}{\sqrt{N^2+1}}\right|\right) + \ln(2)
\end{aligned}
$$

As $N \to \infty$,

$$-\frac{\ln|N^2+1|}{2N^2} \approx -\frac{\ln|N^2|}{2N^2} = -\frac{2\ln|N|}{2N^2} = -\frac{\ln|N|}{N^2}.$$

Since the growth of N^2 outpaces $\ln|N|$, this value tends to 0 (a more formal argument would use L'Hopitals). Also, as $N \to \infty$,

$$\ln\left|\frac{N}{\sqrt{N^2+1}}\right| \approx \ln\left|\frac{N}{\sqrt{N^2}}\right| = \ln\left|\frac{N}{|N|}\right| = \ln|1| = 0.$$

So, the limit evaluates to 0, and we get the value of the integral to be

$$\int_1^{\infty}\int_1^{\infty} \frac{1}{xy(x^2+y^2)}dxdy = \frac{1}{2}\int_1^{\infty} \frac{\ln|y^2+1|}{y^3}dy = \boxed{\frac{\ln(2)}{2}},$$

which means the statement is true for positive integers $m = n = 2$.

2 Finding n

1. $n < 10^9$.

- Puts bounds on n.

2. The last two digits of n in base 2 are 00.

- n is divisible by 4.

3. n, when expressed in base 36, has 5 digits.

- Bounds n, $36^5 \leq n < 36^6$.

4. n is one of the terms of an infinite arithemetic sequence with $a_1 = 1$ and $a_2 = 2024$.

- n is of the form $1 + 2023k$, for nonnegative integer k.

5. Instead of H_3, use hint N_3.

- Disregard the base 36 bounds. Now, $63^4 \leq n < 63^5$.

6. n is divisible by 107.

- $n = 107k$ for some positive integer k.

7. $n \equiv b_{2023} \equiv \prod_{i=1}^{2023} \left(1 + \frac{1}{i}\right)^2 \pmod 9$.

- We can show that $b_k = (k+1)^2$. We have that $\prod_{i=1}^{k} \left(1 + \frac{1}{i}\right)^2 = \left(\prod_{i=1}^{k} \frac{i+1}{i}\right)^2$, and the inner product is a telescoping product that converges to $k+1$. So, $b_k = (k+1)^2$, and we have $n \equiv 2024^2 \equiv (45^2 - 1)^2 \equiv 1 \pmod 9$.

8. $\log_5 n > 8(\log_5 2) + 9$

- Since the exponential function 5^x is strictly increasing (and continuous), we know that if $\log_5 n > 8(\log_5 2) + 9$, then $5^{\log_5 n} > 5^{8(\log_5 2)+9}$. This simplifies, by log rules, to $n > 2^8 \cdot 5^9 = 5 \cdot 10^8$.

9. $n \equiv \sum_{k=1}^{20} k\binom{20}{k} \pmod{19}$

- I claim that $\sum_{i=1}^{k} i\binom{k}{i} = k2^{k-1}$. Using a combinatorical argument, the left side is all the ways to choose any positive number of people from a group of k, then designating one person to lead the chosen. This is equivalent to choosing one person to lead in the beginning, then each of the $k-1$ others are either in the group or not, which can be done in $k2^{k-1}$ ways. This can also be shown algebraically. So, $n \equiv 20 \cdot 2^{19} \pmod{19} \equiv (1)(2 \cdot 2^{18}) \equiv 2 \pmod{19}$. We know $2^{18} \equiv 1 \pmod{19}$ from Fermat's Little Theorem.

10. n is the element of S with the minimum sum of digits.

Since $10^9 > 63^5$ and $5 \cdot 10^8 > 63^4$, we have $5 \cdot 10^8 \leq n < 63^5$. We also know that $n \equiv 0 \pmod 4$, $n \equiv 1 \pmod{2023}$, $n \equiv 1 \pmod 9$, $n \equiv 0 \pmod{107}$, and $n \equiv 2 \pmod{19}$. Since $\gcd(4, 2023, 9, 107, 19) = 1$, by the Chinese Remainder Theorem, these equivalences guarantee a unique $n \pmod{4 \cdot 2023 \cdot 9 \cdot 107 \cdot 19 = 148059324}$, in this case $n \equiv 6645556 \pmod{148059324}$. The possible values of n that satisfy this and the bounds are $598882852, 746942176$, and 895001500. The sum of the digits of each of these are 55, 46, and 28 respectively, so $n = \boxed{895001500}$.